

Experiment No. 5

Aim: Implementation of Fuzzy Membership Function

Theory:

Fuzzy membership functions are fundamental to fuzzy logic and fuzzy set theory. They represent the degree to which a given input belongs to a fuzzy set, with values typically ranging between 0 (completely outside the set) and 1 (completely inside the set). These functions are key components in fuzzy systems, especially for fuzzy inference engines.

Implementing a fuzzy membership function involves defining the shape of the function and calculating the degree of membership for a given input. Common types of fuzzy membership functions include triangular, trapezoidal, Gaussian, and sigmoid functions.

Here's an example of how to implement some common fuzzy membership functions in Python:

1. Triangular Membership Function

The triangular membership function is defined by three parameters: A, B, and C, which represent the Low, Medium, and High points of the triangle.

2. Trapezoidal Membership Function

The trapezoidal membership function is defined by two parameters: D and E, where D and E are the Mid and Wide Range.

3. Gaussian Membership Function

The Gaussian membership function is defined by two parameters: the mean = 5 and the standard deviation $\sigma = 1.5$.

4. Sigmoid Membership Function

The sigmoid membership function is a smooth, S-shaped curve often used in fuzzy system with Sigmoid MF centred at 5 with steepness 1

5. Bell Membership Function

This function computes fuzzy membership values using a generalized bell-shaped membership function. You can also compute this membership function using a fsmf object.

$$f(x; a, b, c) = \frac{1}{1 + \left| \frac{x - c}{a} \right|^{2b}}$$

Input:

Below is an implementation of these common fuzzy membership functions:

Step 1: Importing Necessary Packages

```
import numpy as np
import matplotlib.pyplot as plt
import skfuzzy as fuzz
```

Step 2: Define Universe of Discourse

```
x = np.arange(0, 11, 1)
```

Step 3: Fuzzy Membership Functions

1. Triangular Membership Function

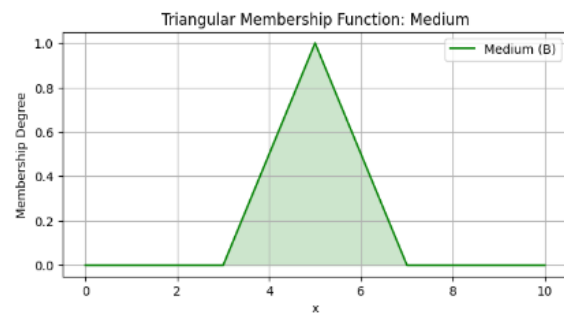
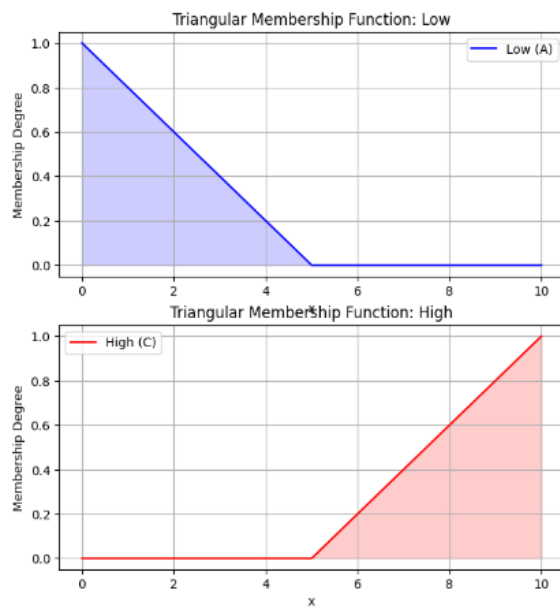
```
[3]: A = fuzz.trimf(x, [0, 0, 5])      # Low
     B = fuzz.trimf(x, [3, 5, 7])      # Medium
     C = fuzz.trimf(x, [5, 10, 10])    # High
```

```
[4]: plt.figure(figsize=(16, 12))

     # Plot Triangular Membership Functions
     plt.subplot(3, 2, 1)
     plt.plot(x, A, label='Low (A)', color='blue')
     plt.fill_between(x, 0, A, color='blue', alpha=0.2)
     plt.title('Triangular Membership Function: Low')
     plt.xlabel('x')
     plt.ylabel('Membership Degree')
     plt.legend()
     plt.grid()

     plt.subplot(3, 2, 2)
     plt.plot(x, B, label='Medium (B)', color='green')
     plt.fill_between(x, 0, B, color='green', alpha=0.2)
     plt.title('Triangular Membership Function: Medium')
     plt.xlabel('x')
     plt.ylabel('Membership Degree')
     plt.legend()
     plt.grid()

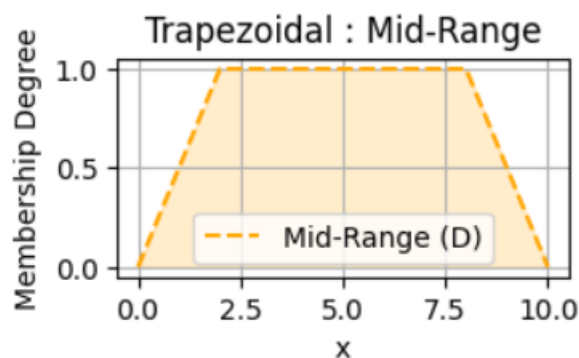
     plt.subplot(3, 2, 3)
     plt.plot(x, C, label='High (C)', color='red')
     plt.fill_between(x, 0, C, color='red', alpha=0.2)
     plt.title('Triangular Membership Function: High')
     plt.xlabel('x')
     plt.ylabel('Membership Degree')
     plt.legend()
     plt.grid()
```



2. Trapezoidal Membership Function

```
[5]: D = fuzz.trapmf(x, [0, 2, 8, 10]) # Mid-Range
     E = fuzz.trapmf(x, [0, 3, 7, 10]) # Wide Range
```

```
[6]: # Plot Trapezoidal Membership Functions
plt.subplot(3, 2, 4)
plt.plot(x, D, label='Mid-Range (D)', color='orange', linestyle='--')
plt.fill_between(x, 0, D, color='orange', alpha=0.2)
plt.title('Trapezoidal : Mid-Range')
plt.xlabel('x')
plt.ylabel('Membership Degree')
plt.legend()
plt.grid()
```



3. Gaussian Membership Function

```
[7]: F = fuzz.gaussmf(x, mean=5, sigma=1.5) # Gaussian centered at 5 with spread 1.
```

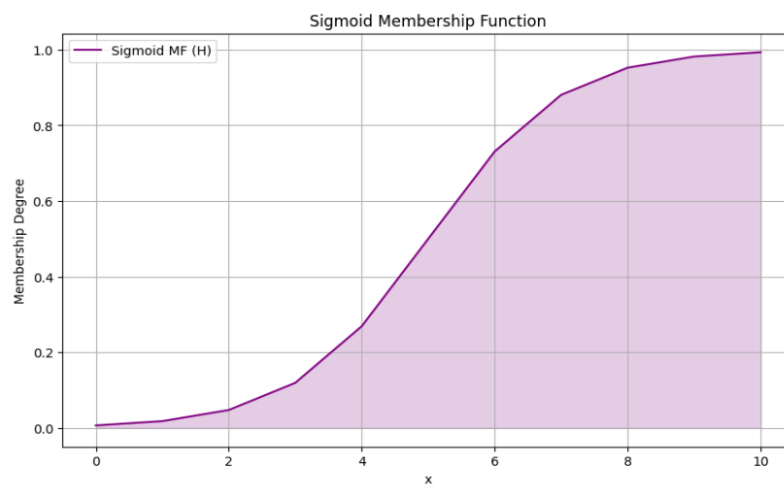
```
[8]: # Plot Gaussian Membership Function
plt.subplot(3, 2, 5)
plt.plot(x, F, label='Gaussian (F)', color='purple')
plt.fill_between(x, 0, F, color='purple', alpha=0.2)
plt.title('Gaussian Membership Function')
plt.xlabel('x')
plt.ylabel('Membership Degree')
plt.legend()
plt.grid()
```



4. Sigmoid Membership Function

```
[11]: H = fuzz.sigmf(x, 5, 1) # Sigmoid MF centered at 5 with steepness 1
```

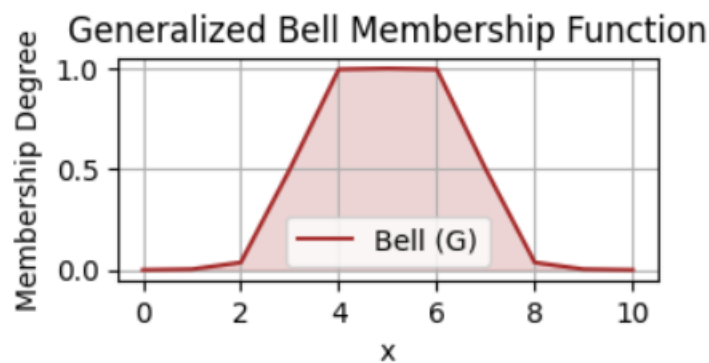
```
[12]: # Visualize the Sigmoid Membership Function
plt.figure(figsize=(10, 6))
plt.plot(x, H, label='Sigmoid MF (H)', color='purple')
plt.fill_between(x, 0, H, color='purple', alpha=0.2)
plt.title('Sigmoid Membership Function')
plt.xlabel('x')
plt.ylabel('Membership Degree')
plt.legend()
plt.grid()
plt.show()
```



5. Bell Membership Function

```
[9]: G = fuzz.gbellmf(x, a=2, b=4, c=5)
```

```
[10]: # Plot Generalized Bell Membership Function
plt.subplot(3, 2, 6)
plt.plot(x, G, label='Bell (G)', color='brown')
plt.fill_between(x, 0, G, color='brown', alpha=0.2)
plt.title('Generalized Bell Membership Function')
plt.xlabel('x')
plt.ylabel('Membership Degree')
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()
```



Conclusion: Thus, we have successfully studied and implemented the Fuzzy Membership Function using Example.